Technical Data Sheet
Revision 2, October 2026

3M™ PELTOR™ X Series Earmuffs

Product description

The 3M™ PELTOR™ X Series Earmuffs are available in multiple design styles, including headband and attachable models suitable for mounting on to carriers. When correctly selected and worn these products can help to reduce exposure to hazardous levels of noise.

These earmuffs feature a colour-coded system that allows quick, simple, and visual identification of the level of protection being worn. The attenuation values start at 27 dB, ranging up to 37 dB in the headband models to meet the needs of a wide range of environments and applications.

Electrically insulated (dielectric) properties:

- Non-conductive (sometimes referred to as “dielectric”) material covers the metal components of the carrier attachment mechanisms.
- These earmuffs have been evaluated at an external laboratory against a modified test method based on EN 397:2012 under dry conditions. The metal parts of the headband are electrically insulated withstanding a voltage up to 1.2 kV.
- The user must determine the overall suitability of this product for the intended application taking into account any hazards other than noise for which this product is tested and approved.



X1A



X2A



X4A



X5A

Key features

- Lightweight design (lightest version 184g)
- The twin headband design not only helps reduce heat buildup, it also provides excellent fit and balance.
- 4-Point, twin headband stainless steel design offers consistent pressure and comfortable fit.
- Electrically insulated headband with no exposed wires helps protect against electrical voltage hazards
- Soft wide foam cushions helps reduce pressure around the ears and improves comfort and wearability
- The cushions and foam inserts, available as Hygiene Kits, are easy to replace if they become worn or damaged, helping extend the life of the product
- Innovative foam earcup inserts and spacer help boost attenuation and protect against low frequencies (X4, X5)
- Despite larger earcups, X5 is lightweight with excellent balance and comfort.
- Easy to understand different colour coding for attenuation values to help select appropriate product for specific application
- ABS plastic cups are rigid, high strength and impact resistant
- Earcups tilt to ensure a customized fit and comfort for individual users.

Colour code and attenuation values

Model	Cup colour band	Headband SNR (dB)
3M™ PELTOR™ Earmuff X1 (X1A)	Green	27
3M™ PELTOR™ Earmuff X2 (X2A)	Yellow	31
3M™ PELTOR™ Earmuff X4 (X4A)	Bright Green	32
3M™ PELTOR™ Earmuff X4 (X4A-OR)	Orange	32
3M™ PELTOR™ Earmuff X5 (X5A)	Black	37

Material listing

Component	Headband
Headband	TPE (Thermoplastic Elastomer), PP, POM (Polyoxymethylene), Stainless steel
Cups	ABS, TPU (Thermoplastic Polyurethane)
Insert (liner)	PU (Polyurethane) foam
Cushions and cushion covers	PVC (Polyvinyl chloride), PU foam
X4, X5 Spacer	ABS (Acrylonitrile butadiene styrene)

Standards and approvals

Hereby, 3M Svenska AB declares that the product is in compliance with appropriate Directives or Regulations to fulfil the requirements for the CE marking. The full text of the Declaration of Conformity is available at the following internet address: www.3M.com/PELTOR/DOC.

A copy of the Declaration of Conformity and additional information required in the Directives or Regulations can also be obtained by contacting 3M in the country of purchase.

Personal Safety Division User Instruction Selector is available at; 3M.com/userinstructions

Please note this site is subject to updates and may not have all user instructions available. If you cannot find your online user instruction, please reference the printed user instruction that was shipped with the product.

3M strongly recommends personal fit testing of hearing protectors. Research suggests that users may receive less noise reduction than indicated by the attenuation label value(s) on the packaging due to variation in fit, fitting skill, and motivation of the user. Refer to applicable regulations and guidance on how to adjust attenuation label value(s). In the absence of applicable regulations, it is recommended that the attenuation label value(s) be reduced to better estimate typical protection.

Accessories/replacement

The cushions and inserts on the 3M™ PELTOR™ X Series earmuffs can be replaced with the hygiene kits listed below to maintain consistent protection, hygiene, and comfort.

Hygiene Kits

Earmuff model	Hygiene kit
3M™ PELTOR™ Earmuff X1	HYX1
3M™ PELTOR™ Earmuff X2	HYX2
3M™ PELTOR™ Earmuff X4	HYX4
3M™ PELTOR™ Earmuff X5	HYX5

In addition, HY100 sweat pads are also available that can be placed on the cushions to help absorb moisture and sweat.

Attenuation values and weights

3M™ PELTOR™ X1A Earmuff								**EN 352-1:2020**				
	Frequency (Hz) *f*							**H**	**M**	**L**	**SNR**	**Weight**
	125	**250**	**500**	**1000**	**2000**	**4000**	**8000**					
Mf (dB)	11.9	15.4	24.5	34.3	32.8	37.4	37.4	34.3	25.8	17.8	28.2	184 g
SD (dB)	2.0	2.6	2.6	2.3	3.3	2.5	3.8	2.1	1.8	1.9	1.7	
APVf (dB)	9.9	12.8	21.9	32.0	29.5	34.9	33.6	32	24	16	27	

3M™ PELTOR™ X2A Earmuff								**EN 352-1:2020**				
	Frequency (Hz) *f*							**H**	**M**	**L**	**SNR**	**Weight**
	125	**250**	**500**	**1000**	**2000**	**4000**	**8000**					
Mf (dB)	14.1	22.2	31.1	39.7	36.6	37.0	37.9	37.4	31.3	22.2	33.0	220 g
SD (dB)	2.2	2.1	2.7	3.2	3.2	3.7	3.4	2.6	1.8	1.9	1.8	
APVf (dB)	11.9	20.1	28.4	36.6	33.4	33.3	34.5	35	30	20	31	

3M™ PELTOR™ X4A Earmuff								**EN 352-1:2020**				
	Frequency (Hz) *f*							**H**	**M**	**L**	**SNR**	**Weight**
	125	**250**	**500**	**1000**	**2000**	**4000**	**8000**					
Mf (dB)	17.8	22.1	30.6	39.5	37.3	43.8	42.1	39.3	32.1	24.2	34.3	234 g
SD (dB)	2.3	2.5	1.8	2.9	4.1	2.8	4.0	2.9	1.9	2.2	1.9	
APVf (dB)	15.5	19.6	28.8	36.6	33.2	41.0	38.1	36	30	22	32	

3M™ PELTOR™ X5A Earmuff								EN 352-1:2020				
	Frequency (Hz) *f*							H	M	L	SNR	Weight
	125	250	500	1000	2000	4000	8000					
Mf (dB)	22.3	28.8	39.7	44.2	39.8	43.0	40.2	40.8	37.8	29.9	39.0	351 g
SD (dB)	2.4	2.4	2.7	3.4	4.6	2.8	2.9	3.2	1.7	2.1	1.9	
APVf (dB)	19.9	26.4	37.0	40.8	35.2	40.2	37.3	38	36	28	37	

Attenuation table key:

f = Test frequency

Mf = Mean attenuation value

SD = Standard deviation

APVf (Mf - SD) = Assumed Protection Value

H = High-frequency attenuation value (predicted noise level reduction for noise with LC – LA = -2dB)

M = Medium-frequency attenuation value (predicted noise level reduction for noise with LC – LA = +2dB)

L = Low-frequency attenuation value (predicted noise level reduction for noise with LC – LA = +10dB)

SNR = Single Number Rating (the value that is subtracted from the measured C-weighted sound pressure level, LC in order to estimate the effective A-weighted sound pressure level inside the ear).

Important Notice

Product selection and use: Many factors beyond 3M's control and uniquely within user's knowledge and control can affect the use and performance of a 3M product in a particular application. As a result, customer is solely responsible for evaluating the product and determining whether it is appropriate and suitable for customer's application, including conducting a workplace hazard assessment and reviewing all applicable national and/or European regulations and standards. Failure to properly evaluate, select, and use a 3M product in accordance with all applicable instructions and with appropriate safety equipment, or to meet all applicable safety regulations, may result in injury, sickness, death, and/or harm to property.

Warranty, limited remedy, and disclaimer: A limitation of liability applies to the 3M product(s). For warranty statement and limitation of liability, refer to your supply agreement or the 3M terms and conditions of sale.

3M industrial and occupational products are intended, labeled, and packaged for sale to trained industrial and occupational customers for workplace use.

3M United Kingdom Ltd.
Building 1 050
Eskdale Road
Winnersh
RG41 5TS

3M Ireland Limited
Ella House, 39-43 Merrion Square,
Dublin 2, D02 NP96, Ireland.

3M PSD products are occupational use only.

© 2026, 3M. All rights reserved.
3M and PELTOR are trademarks of 3M.
All rights reserved.

Please recycle.